

Codenames: A Knowledge Graph Approach to Adaptive Clue Generation

Katie Baakkonen, Aasritha Gopalam, Kanagha Sivakumar

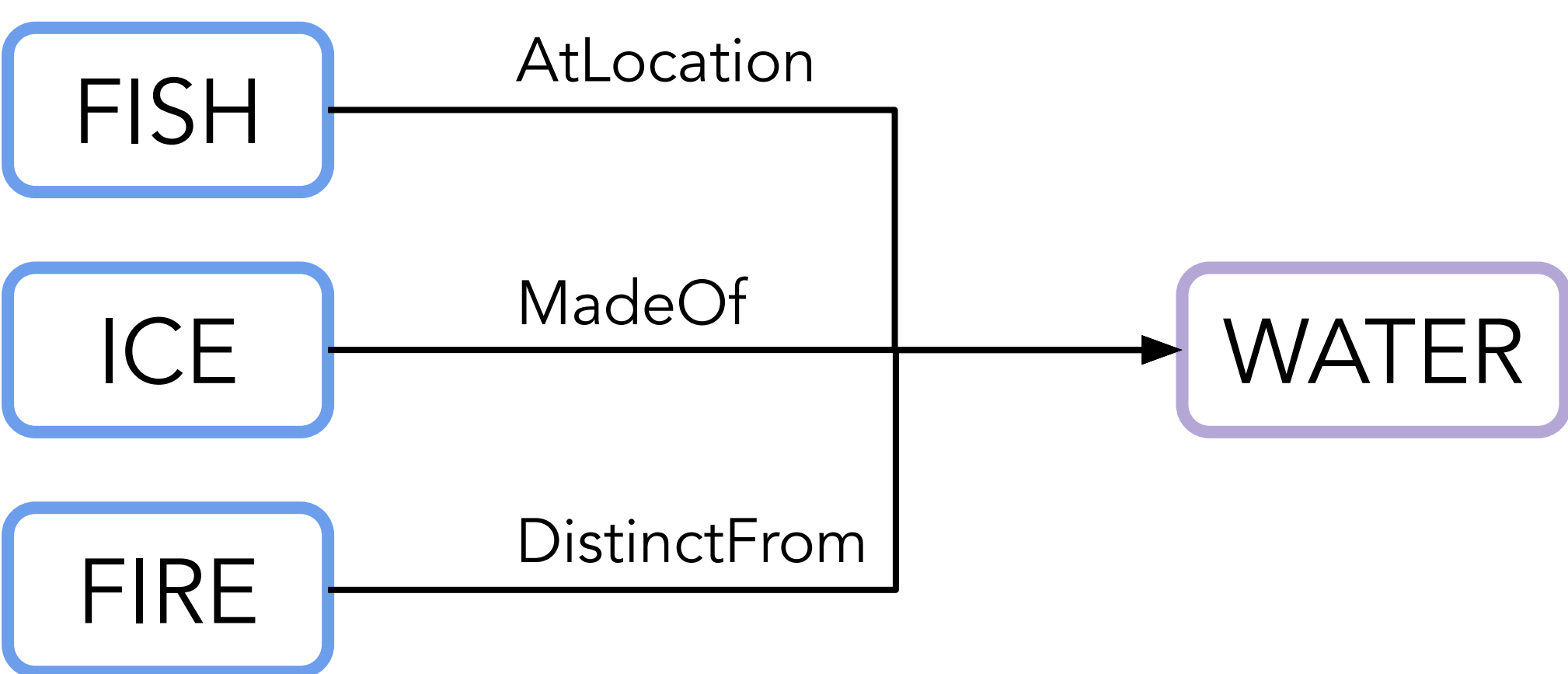
Codenames: the Game

Codenames is a board game that requires players to guess certain **words** on a board while avoiding guessing their opponents' words. They make their guesses based off a **clue** word given by their team's **spymaster**. The clue should relate a group of words together. The clue must also avoid words that are in the other team's list.

An example clue is:
WATER
Target words:
FISH, ICE, FIRE

ConceptNet

We used ConceptNet to build the knowledge graph that is searched to find clues. ConceptNet has different relations that are used to connect words together.



Results

Board Reveal: At game end, all 25 words are revealed with full team color assignments — red, blue, neutral, assassin. Guessed words appear struck-through.

OLYMPUS	BRUSH	WAR	AMBULANCE	WALL
TEACHER	WELL	ROOT	NUT	CONTRACT
FISH	CHOCOLATE	DRAGON	CAST	SOLDIER
LEMON	LUCK	ANGEL	DROP	BEAR
HAM	TABLET	CAPITAL	LOCK	LINK

Decision Tree: Each player sees a turn-by-turn trace of every clue given, which relations connected it to the targets, and weight before → after each update.

TURN 2 clue FASTENER (IsA×2)
├─● LINK guessed ✓ [IsA 0.097 → 0.102 ↑ rewarded]
└─● LOCK guessed ✓ [IsA 0.097 → 0.102 ↑ rewarded]

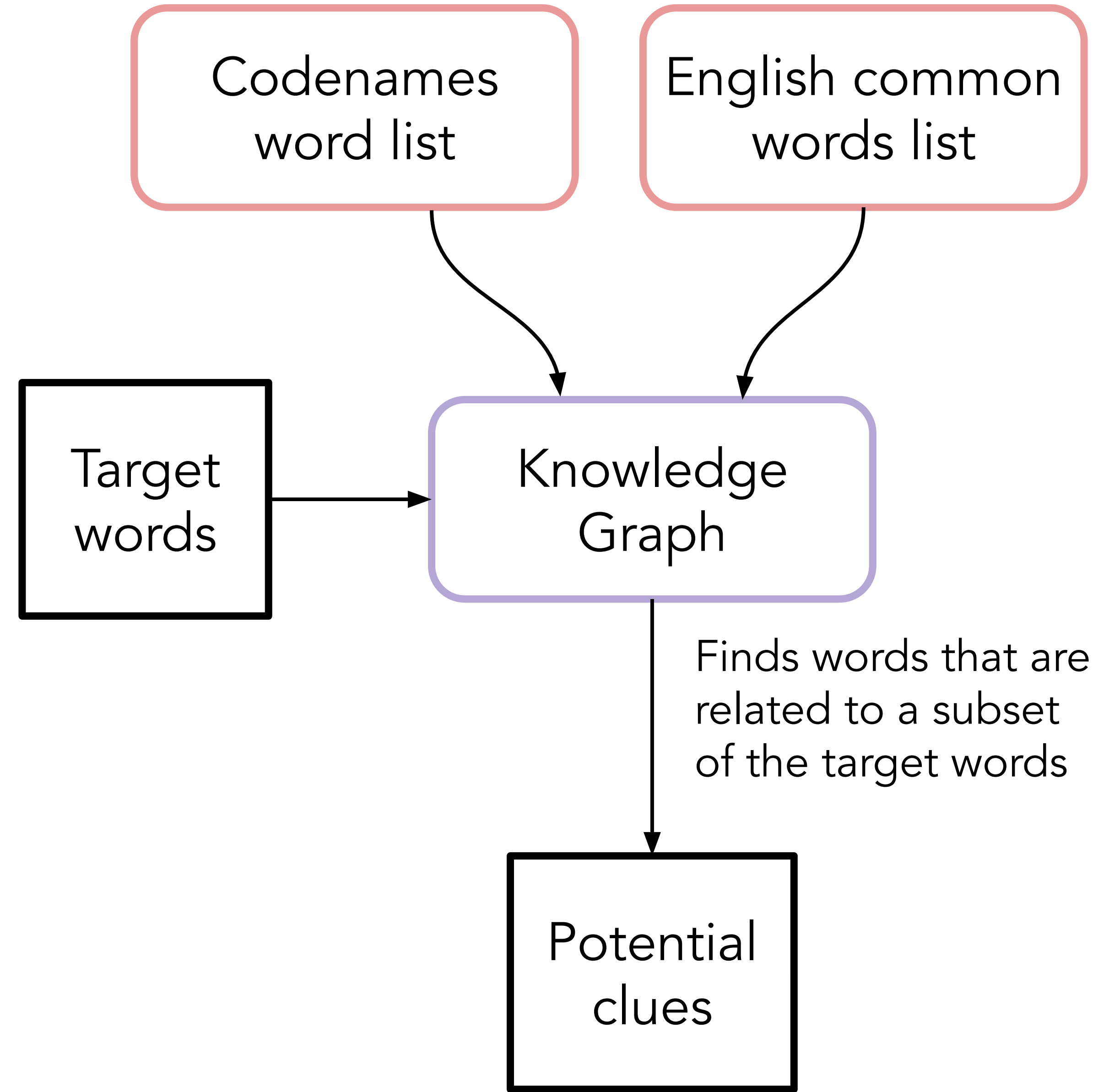
TURN 3 clue FIGHT (IsA×2)
├─○ BRUSH missed ✗ [IsA 0.102 → 0.097 ↓ decayed]
└─○ WAR missed ✗ [IsA 0.102 → 0.097 ↓ decayed]

TURN 4 clue ADVISABLE (SimilarTo×1) ⇐ pivoted relation
└─● WELL guessed ✓ [SimilarTo 0.857 → 0.900 ↑ rewarded]

Clue Profile: Hit rate per relation type across all turns, showing which relation types each player responds well to vs. struggles with.

- ✓ You respond well to:
 - AtLocation 100% hit rate (2/2 words guessed)
e.g. clue "REFRIGERATOR" → you found FISH
 - SimilarTo 100% hit rate (1/1 words guessed)
e.g. clue "ADVISABLE" → you found WELL
 - Antonym 100% hit rate (1/1 words guessed)
e.g. clue "PICK" → you found DROP
- ✗ You struggle with:
 - ▨ PartOf 0% hit rate (0/1 words guessed)
e.g. clue "VENEZUELA" → missed ANGEL
- ~ Mixed results with:
 - ▨ IsA 50% hit rate (2/4 words guessed)

Solution



Dynamic Clue-Giving

For each user, the spymaster learns which relations they typically guess correctly and weighs those relations more heavily so the clues it gives are more likely to be understood.

Representation: ConceptNet edges weighted by relation type with per-relation normalization (IsA $\alpha=0$, UsedFor $\alpha=0.8$)

Reasoning: Soft penalty scoring, bad word connections subtract from a clue's score rather than hard-rejecting it. Coverage bonus (2×per extra target) forces multi-word clues.

Learning: On a miss → weight × 0.95. On a hit → weight × 1.05. The spymaster adapts to each player over time. When it abandons a failing relation type entirely after a miss, this is flagged as a pivot.